

FILTER DESIGN TEST CASE

Lattice Filter - Band 20 (800 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting Band 20 (800 MHz) for Automotive V2X Communication. The filter is designed on AlN on SiC substrate with a target center frequency of 3652.5 MHz and bandwidth of 221.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: Band 20 (800 MHz)
- Application: Automotive V2X Communication
- Substrate: AlN on SiC
- Package: WLP (Wafer Level Package)
- Design Tool: PathWave EM Design
- Design Center: Andover HQ Lab

This specification applies to all variants of the Lattice Filter design targeting Band 20 (800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3652.5 MHz
Bandwidth (3 dB):	221.3 MHz
Insertion Loss Target:	≤ 2.3 dB
Return Loss Target:	≥ 15.0 dB
Out-of-Band Rejection:	≥ 22.0 dB
Isolation (Duplexer):	≥ 23.0 dB
Q Factor:	8402
Temperature Coefficient:	-16.2 ppm/°C
Die Size:	2.4 x 0.9 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Mo/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Lattice Filter template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 3652.5 MHz, BW 221.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on SiC).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10958 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.3 dB
- Return loss in passband shall be >= 15.0 dB
- Out-of-band rejection at +/- 332 MHz offset >= 22.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 7.4 MHz
- Power handling shall be >= +35 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 88%

6. TEST RESULTS

Measurement Date: 2024-03-10
Devices Tested: 66
Insertion Loss (meas): 2.05 dB
Return Loss (meas): 17.2 dB
Rejection (meas): 25.0 dB
Isolation (meas): 28.3 dB
Q Factor (meas): 8402
Group Delay Variation: 4.7 ns
Power Handling: +36 dBm
Die Yield: 95.2%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated acceptable performance for Band 20 (800 MHz).
- b) Insertion loss met target specification.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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